

SonicGen: Audio-Driven Player DNA & Soundtrack AI

A Multimodal AI Pipeline for Personalized Gaming Experiences

By Prathyu Adari

CS 5588 – Week 14 Hands-On Challenge

Problem Description

The Problem:

Modern video games use static soundtracks that do not adapt to individual player behaviors.

Profiling player behavior typically relies on raw telemetry data (K/D ratio, movement speed) rather than emotional or communication context.

The Solution:

An AI system that ingests a player's voice communications, extracts an '8-Axis Gamer DNA' profile, and dynamically generates custom background music to match their emotional state and playstyle.

Business Value & Why It Matters

Enhanced Immersion:

Players stay in the 'flow state' longer when audio matches their vibe and current situational intensity.

Player Retention:

Personalized gaming experiences dramatically increase lifetime value (LTV) and session length.

Esports & Streaming:

Streamers get auto-generated, copyright-free background music tailored to how intensely they are playing or speaking.

Dataset & Inputs Used

Speech Input:

Simulated 10-second audio clips of player voice chat.

Example A: 'Push mid, they're weak! Flank them!'

Example B: 'Let's just gather resources and build a base quietly.'

Music Output:

AI-generated 10-second soundtracks aligned with the speech input.

Pipeline Architecture & Models Used

Phase 1: Speech-to-Text

openai/whisper-small transcribes the player's voice chat.

Phase 2: AI Agent / DNA Extraction

An LLM agent analyzes the text to determine the '8-Axis Gamer DNA' (Aggression, Tactical, Stealth, Chaos, etc.) and outputs a music prompt.

Phase 3: Music Generation

facebook/musicgen-small takes the generated prompt and creates the custom soundtrack.

Prompt & Input Engineering

Speech Prompting:

Tested clean vs. noisy gaming audio to evaluate Whisper's resilience to background game noise.

MusicGen Prompt Engineering:

Mapped specific DNA traits to precise audio parameters.

High Aggression DNA -> 'Heavy bass, fast tempo 140bpm, aggressive synth, cyberpunk'

High Stealth/Calm DNA -> 'Ambient, slow tempo, tense strings, quiet atmospheric'

Evaluation (Baseline vs. Improved)

Baseline:

Feeding MusicGen generic prompts like 'Video game music.'

Improved Setting:

Injecting our 8-Axis DNA prompt structure (e.g., 'Aggressive cyberpunk battle music, 140bpm, heavy percussion').

Metrics Tracked:

Whisper WER (Word Error Rate): Evaluated accuracy on gaming slang (e.g., 'gank', 'DPS').

Music Prompt Alignment: Evaluated if the generated audio matched the mood of the voice chat.

Results & Insights

Finding 1:

Whisper struggles slightly with gaming-specific slang without fine-tuning, but successfully captures the general sentiment and intent perfectly.

Finding 2:

MusicGen responds remarkably well to genre and BPM (Beats Per Minute) tags triggered by the Player DNA profile.

Conclusion:

Multimodal chaining (Speech -> Text -> Music) creates highly coherent, real-time audio personalization in games.

Limitations & Future Work

Limitations:

Real-time generation latency. MusicGen takes a few seconds to generate audio, which delays instantaneous in-game adaptation.

Future Work:

Integrate AudioLDM2 to generate specific sound effects (e.g., custom gunshot sounds or footsteps) based on the same Player DNA.

Model distillation for edge deployment to reduce latency.

Links & Disclosure

GitHub Repository:

<https://github.com/PrathyushaRagavAdari/GamerZone->

AI Tools Disclosure:

Models: Hugging Face (Whisper, MusicGen)

Compute: Google Colab

Assistance: AI-assisted structuring for the Python pipeline and boilerplate text generation.

**Thank
you!**